

Sadhasivam Perichi

Email: sadhasivamperichi@gmail.com

LinkedIn: linkedin.com/sadhasivam-perichi-160449255/

Mobile: +91-9361215021

Github: github.com/smily-sadha



EDUCATION

- **Sona College of Technology** Salem, India
Bachelor of Engineering - Artificial Intelligence & Machine Learning CGPA: 7.5 2022 – 2026
- **IVL Matric HR. Sec. School** Dharmapuri, India
HSC; 81.5% 2021 – 2022
- **IVL Matric HR. Sec. School** Dharmapuri, India
SSLC; 93.5% 2019 – 2020

INTERNSHIP EXPERIENCE

- **Spacemarshal.ai: AI Engineer Intern** July 2025 – Present
Built agentic AI voice agents for real-world call workflows including cold-calling automation, hospital appointment booking, delivery coordination, and AI-driven lead generation.
Developed a Retrieval-Augmented Generation (RAG) system for hospital SOP assistance using LangChain.
Implemented document ingestion, chunking, embedding, and retrieval pipelines for knowledge-grounded responses.
Integrated real-time speech processing using Deepgram, Pipecat, and LiveKit.
Implemented authentication and session management using Clerk.

PROJECTS

- **Online Voting System with Face Recognition:**
Engineered a secure online voting platform leveraging facial biometrics for real-time voter authentication. Designed a face verification pipeline using OpenCV-based preprocessing and Single Shot Detection for face localization, and implemented identity validation logic to prevent duplicate voting attempts. Developed backend logic for vote validation and persistence using MongoDB, along with interactive web workflows for voter registration and ballot submission.
Tech: Python, OpenCV, MongoDB, Single Shot Detection
- **Plant Leaf Disease Detection System:**
Built a CNN-based image classification model for plant disease detection, including dataset preprocessing and normalization for model stability. Integrated the trained model into a Flask web application for inference and designed prediction pipelines handling image validation and preprocessing.
Tech: Python, TensorFlow/Keras, OpenCV, Flask
- **Online Job Application Web Portal:**
Developed a full-stack job application system supporting authentication and job workflows. Implemented RESTful APIs using Node.js and Express, built responsive UI components using React, and managed persistent storage using MongoDB.
Tech: React, Node.js, Express, MongoDB
- **Online Attendance System using Face Recognition:**
Automated attendance tracking using live webcam input and CNN-based feature extraction. Implemented face recognition using SVM classifiers, designed preprocessing pipelines for illumination variance handling, and optimized frame processing for stable recognition behavior.
Tech: Python, OpenCV, CNN, SVM

SKILLS SUMMARY

- **Programming:** Java, Python
- **Machine Learning:** scikit-learn, Feature Engineering, Model Validation
- **Deep Learning:** PyTorch, TensorFlow/Keras, CNN, GAN
- **Data Analytics:** Python (NumPy, Pandas, Matplotlib, Seaborn), Excel, Power BI, Tableau
- **Web Development:** HTML, CSS, ReactJS, Node.js, Express
- **Databases:** MongoDB (Atlas), MySQL Workbench

AREA OF INTEREST

- AI Engineer
- Data Analytics
- Full-Stack Web Development
- AI Agent Developer

CERTIFICATION

- NoviTech – Data Analyst, Artificial Intelligence, Machine Learning, Full Stack Developer
- Coursera – Data Structures
- Infosys – Foundation of Java
- Guvi – Basics of Python
- NPTEL – Design & Implementation of Human Computer Interface

ACHIEVEMENTS

- Secured **1st place** in Football (2022), Sona College of Technology
- Secured **3rd place** in UiPath Hackathon (2024), Sona College of Technology